

GCP / AZURE /AWS GUIDE :

Cloud Services Comparison — GCP vs Azure vs AWS

Category	GCP	Azure	AWS
Compute Services	Compute Engine / App Engine / Cloud Functions / Cloud Run / GKE	Virtual Machines / App Service / Azure Functions / Container Apps / AKS	EC2 / Elastic Beanstalk / Lambda / App Runner / EKS
Storage Services	Cloud Storage / Persistent Disk / Filestore / Archive Storage	Blob Storage / Managed/Files / Archive Tier	S3 / EBS / EFS / Glacier
Database Services	Cloud SQL / Spanner / Firestore / Bigtable / Memorystore	SQL Database / Cosmos DB / Table Storage / Cache for Redis	RDS / DynamoDB / Aurora / ElastiCache
Networking	VPC / Cloud Load Balancing / Cloud CDN / Cloud DNS / Cloud VPN / Interconnect	VNet / Load Balancer + App Gateway / Azure CDN / DNS / VPN Gateway / ExpressRoute	VPC / ELB + ALB + NLB / CloudFront / Route 53 / VPN Gateway / Direct Connect
AI / ML Services	Vertex AI / AutoML / Vision AI / Natural Language / Speech / Translation	Azure Machine Learning / AutoML / Cognitive Services (Vision, Text, Speech)	SageMaker / Rekognition / Comprehend / Polly / Translate

Category	GCP	Azure	AWS
Analytics & Big Data	BigQuery / Dataflow / Dataproc / Pub/Sub / Data Fusion	Synapse Analytics / Data Factory / Stream Analytics / Event Hubs	Redshift / Athena / Kinesis / Glue / EMR / SNS / SQS
Security & Identity	Cloud IAM / Cloud KMS / Secret Manager / Armor / Security Command Center	Azure AD (Entra ID) / Key Vault / Defender for Cloud / DDoS & WAF	IAM / KMS / Secrets Manager / WAF / Shield / Security Hub
Monitoring & Management	Cloud Monitoring / Cloud Logging / Cloud Trace / Deployment Manager / Audit Logs	Azure Monitor / Log Analytics / Application Insights / Resource Manager / Activity Log	CloudWatch / X-Ray / CloudFormation / CloudTrail
Containers & Orchestration	Artifact Registry / Cloud Build / GKE Autopilot	Container Registry / AKS Automatic / Azure DevOps	ECR / CodeBuild / EKS / Fargate
Serverless & Integration	Cloud Functions / Cloud Scheduler / Workflows / Eventarc / Cloud Tasks	Azure Functions / Logic Apps / Event Grid / Automation	Lambda / Step Functions / EventBridge / SQS
Developer Tools	Cloud SDK / Console / Cloud Shell / Cloud Code / Source Repos / Cloud Build	Azure CLI / Portal / Cloud Shell / VS Code Extensions / Azure Repos / Pipelines	AWS CLI / Management Console / CloudShell / Toolkit / CodeCommit / CodePipeline

Feature	Google Cloud TTS	Azure Speech Service (TTS)	Amazon Polly
Launch Year	2018	2019	2016
Languages Supported	40+	140+ (Best)	30+
Voices Available	220+	400+ (Largest)	60+
Best Model	Neural2 (balanced cost/quality)	Neural HD (highest quality)	Neural Joanna/Matthew
Voice Cloning	✗ None	✓ Personal Voice (1-min)	✗ None
Custom Voice	✓ Available	✓ Custom Neural Voice	✓ Brand Voice
Emotion & Style Control	Limited	✓ Advanced Styles	✓ Speaking Styles
Multilingual Voice	Limited	✓ Multilingual Neural	Limited
Real-Time Streaming	✓ Yes	✓ Yes	✓ Yes
Language Auto-Detection	✗ No	✓ Yes	✗ No

Best Model Types by Provider

Provider	Model Type	Example Name	Best For
GCP	Neural2	en-US-Neural2-J	Natural, versatile speech
	Studio	en-US-Studio-B	Audiobooks, long narration
Azure	Neural HD	en-US-AvaMultilingualNeural	High-end, multilingual
	Custom Neural Voice	Custom brand voice	Unique branding
	Personal Voice	—	Accessibility & personalization
AWS	Neural	Joanna (Neural)	Conversational, podcasts
	Long-form Neural	Matthew Long-form	Audiobooks, e-learning
	Newscaster	Matthew Newscaster	News reading

Text-to-Speech Pricing Comparison (per 1M characters)

Quality Level	GCP	Azure	AWS
Standard	\$4	—	\$4

Quality Level	GCP	Azure	AWS
Neural / Premium	\$16	\$15 (Best Value)	\$16
HD / Long-form	\$100	\$30 (Lowest)	\$100
Custom Voice	Contact Sales	\$6–\$40 + training	Contact Sales

 Quick Decision Summary

Category	Winner	Reason
Best Overall Quality	★ Azure (Neural HD)	Natural, emotional, multilingual
Best Value	Azure Neural (\$15/1M)	Lower cost, broad language support
Best for Audiobooks	GCP Studio / AWS Long-form	Optimized for long-form speech
Best Customization	Azure Custom Neural Voice	Unique brand voice creation
Best Multilingual	Azure	140+ languages & multilingual models

Category	Winner	Reason
Most Mature Service	AWS Polly	Oldest, stable, widely integrated
Best Innovation	GCP Neural2 / Azure Personal Voice	Cutting-edge realism & cloning
Easiest Integration	AWS	Seamless with AWS ecosystem

Summary Highlights

- **Azure** dominates in **languages, customization, and realism** — best for enterprise or global apps.
- **GCP** shines in **long-form, audiobook-quality narration** with **Neural2** and **Studio** models.
- **AWS Polly** remains a **reliable, cost-effective choice** with good SSML and news-style voices.
- For **developers**, all three provide similar **API simplicity** and **SDK support** across platforms.